

Round 2 NW-Sweep: Evaluating the Design Hypotheses Against Real Data

FastMSPEC project report

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Abstract

Round 2 (‘docs/notebook5_revamp_progress.md’'s "Round 2 design" section) ran FastMSpec at 5 candidate bandwidths per pair for a 60-pair, distance-stratified subset (300 work units total), built specifically to test four hypotheses that motivated the sweep's design in the first place. This report states each hypothesis plainly, evaluates it against the real 300-point dataset, and gives an explicit verdict for each – including one result (H2) that did *not* come out the way a naive reading of the theory doc would predict, reported as found rather than smoothed over. Standalone reference, independent of the conversation that produced it.

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1 The Four Hypotheses, Stated for the Record

Recalled from the session that designed Round 2; stated here explicitly so any misremembering can be corrected before the analysis below is trusted.

H1 — Normalized bandwidth is the right variable. A pair's own $NW_{\text{high}}(r) \approx N_{\text{cmin}}/(4r)$ (the resolution-bandwidth ceiling, computed per pair from its distance and local phase velocity) is the correct quantity to normalize candidate bandwidths against – i.e., convergence/quality should behave as a consistent function of $NW/NW_{\text{high}}(r)$ across pairs of very different absolute distance, not depend on absolute NW in a way that differs pair to pair.

H2 — An interior MSE optimum exists. Per the bias-variance framework in `stage5_bandwidth_theory.tex` (following Haley & Anitescu 2017), quality should *peak* at some bandwidth strictly between “as narrow as possible” and the resolution ceiling – not simply keep improving all the way to the ceiling. Haley & Anitescu’s own Lorentzian example landed at $NW = 14.5$ against a much larger ceiling, motivating the expectation of an interior peak here too.

H3 — An existence threshold produces a cliff for far pairs. For sufficiently large distance r (hence small $NW_{\text{high}}(r)$), no candidate bandwidth in a reasonable range should produce reliable convergence at all – a qualitatively different, complete-failure pattern for far pairs, distinct from the gradual bias-variance degradation expected for near pairs.

H4 — A hard numerical floor exists below $NW \approx 3$. Discovered mid-round, not originally hypothesized, but directly testable once found: **FastMultitaper**’s taper count K hits exactly zero below $NW \approx 3$ (this project’s **cutoff**), a structurally different failure mode from H2/H3’s statistical ones, expected to concentrate at low fractions and, because $NW_{\text{high}}(r)$ shrinks with r , disproportionately on far pairs at any *fixed* fraction.

2 Data

300 work units: 60 pairs (15 per distance quartile, using Round 1’s own quartile boundaries) x 5 bandwidth fractions each, $NW/NW_{\text{high}}(r) \in \{1.5, 1.0, 0.7, 0.45, 0.25\}$. Breakdown: **60/300 converged (20%)**, **65/300 (21.7%) hit the H4 floor** ($K = 0$, a clear, diagnosable error after the same-day fix – see `docs/notebook5_revamp_progress.md`’s 2026-09-04 log entry), **175/300** are genuine convergence/non-convergence outcomes at numerically valid bandwidths. All numbers below are computed directly from `data/results/dispcurve_quality/round2_sweep/` (300 files) and `round2_subset.csv`.

3 H1: Is fraction-of- NW_{high} the right normalization?

Quartile	$NW/NW_{\text{high}} = 1.5$	1.0	0.7	0.45	0.25
1 (nearest)	12/15 (80%)	10/15 (67%)	9/15 (60%)	7/15 (47%)	3/15 (20%)
2	8/15 (53%)	4/15 (27%)	2/15 (13%)	0/15 (0%)	0/15 (0%)
3	2/15 (13%)	3/15 (20%)	0/15 (0%)	0/15 (0%)	0/15 (0%)
4 (farthest)	0/15 (0%)	0/15 (0%)	0/15 (0%)	0/15 (0%)	0/15 (0%)

Table 1: Convergence count/rate by quartile x fraction (raw, $K = 0$ rows included as non-converged).

Verdict: partially supported, with an important qualification. Quartiles 1–3 all show the same *direction* of dependence on fraction (convergence declines as fraction shrinks), consistent with $NW/NW_{\text{high}}(r)$ being a meaningful normalized variable rather than an artifact of absolute distance. But quartile 4 doesn’t participate in this pattern at all – it is uniformly zero across every fraction tested, including $1.5\times$ its own ceiling. Normalizing by $NW_{\text{high}}(r)$ makes quartiles 1–3 comparable to each other; it does not rescue quartile 4, whose true operating point (if one exists at all for this window length) appears to sit entirely outside the tested range. This is exactly what H3 predicts, and is evaluated properly there rather than being treated as a failure of H1.

4 H2: Is there an interior optimum bandwidth?

	1.5	1.0	0.7	0.45	0.25
Convergence rate, raw (n=60 each)	37%	28%	18%	12%	5%
Convergence rate, excl. $K = 0$ rows	37%	28%	21%	17%	14%
Mean <code>bad_quality_fraction</code> (converged only)	0.634	0.651	0.663	0.666	0.800
Mean <code>freq_coverage_fraction</code> (converged only)	0.540	0.486	0.476	0.358	0.366

Table 2: Convergence rate and quality scalars vs. fraction, aggregated across all quartiles.

Verdict: not supported within the tested range $[0.25, 1.5] \times NW_{\text{high}}$. Every measure – raw convergence, $K = 0$ -excluded convergence, and both quality scalars among converged units – moves monotonically in the same direction: wider is better, all the way out to $1.5\times$ the nominal resolution ceiling, with no sign of turning over. This is not the pattern H2 predicted. Two honest readings, not adjudicated here:

1. The true interior optimum lies *above* $1.5 \times NW_{\text{high}}$ – i.e., the resolution ceiling itself is more conservative than the point where bias actually starts dominating, and the sweep simply didn’t reach far enough to see the turnover.
2. In this regime, variance (favoring wider bandwidth) dominates the smooth bias term badly enough, or other project-specific effects (picker mechanics, real non-stationarity per `stage5_bandwidth_theory.tex`’s Section 5.2) swamp it enough, that no interior optimum is visible at this sample size even if one exists in principle.

Distinguishing these needs a wider sweep (fractions above 1.5) before H2 can be considered settled either way – reported as an open question, not resolved by extrapolation.

5 H3: Does a real existence-threshold cliff appear for far pairs?

Verdict: strongly supported, and more starkly than the qualitative prediction. Quartile 4’s 15 pairs (individually, not just in aggregate) show *zero* converged units across all 75 attempts (5 fractions x 15 pairs), including at $1.5\times$ their own already-small $NW_{\text{high}}(r)$ (range 2.30–5.08, all pairs individually listed in the underlying data). This is not a gradual decline – it’s a complete absence of convergence across the entire tested bandwidth range for every single far pair, exactly the qualitative “no valid window exists” signature H3 predicted, and clearer than a marginal cliff would have been.

6 H4: The $K = 0$ numerical floor

Quartile 1	Quartile 2	Quartile 3	Quartile 4
1/75 (1%)	9/75 (12%)	17/75 (23%)	38/75 (51%)

Table 3: $K = 0$ -floor rate by quartile (out of all 75 sweep points per quartile).

Verdict: confirmed, with a clean, mechanistically-explained gradient. The $K = 0$ floor rate rises monotonically from 1% (nearest quartile) to 51% (farthest) – exactly what the mechanism predicts: since $NW_{\text{high}}(r)$ shrinks with distance, the *same* fraction multiplier produces

a smaller absolute NW for farther pairs, crossing the $NW \approx 3$ floor at a higher fraction. By quartile 4, over half of all sweep points are not merely statistically unreliable but numerically undefined. This is a hard, structural limit, categorically different from H2/H3’s statistical ones – restated from `docs/notebook5_revamp_progress.md`: it must not be conflated with the soft, bias-variance-driven lower bound Stage 5 is deriving.

7 Synthesis and Implications for Stage 5

- **H1 and H3 together give a clean, load-bearing result:** normalized bandwidth ($NW/NW_{\text{high}}(r)$) is the right variable for pairs where a viable window exists at all (quartiles 1–3), and the existence-threshold theory correctly predicts *which* pairs don’t have one (quartile 4, confirmed with zero exceptions). This validates the core machinery `round2_prep.py` was built on, not just the sweep’s raw convergence numbers.
- **H2’s non-result is the most actionable finding for Stage 5’s next step:** the bias–variance MSE framework predicts an interior optimum, and this sweep didn’t find one – not because it was falsified, but because the tested range may not have reached it. Before trusting `stage5_bandwidth_theory.tex`’s Eq. 16 recipe as a complete answer, extend the sweep’s upper fraction (e.g. to $2.0\text{--}3.0 \times NW_{\text{high}}$) on a subset of quartile-1 pairs specifically, where a real signal is most likely to be visible against the noise floor.
- **H4 is now a known, permanent constraint on any future sweep design:** any bandwidth search must stay above $NW \approx 3$ or expect the same clean failure mode – already fixed to fail informatively rather than silently, but still worth excluding from statistical fits by construction (e.g. floor the fraction grid’s minimum well above wherever $NW \approx 3$ falls for the pairs being tested) rather than discovered again.

8 Caveats

- Sample sizes are modest – 15 pairs per quartile, and after excluding the $K = 0$ floor, as few as 22 valid units at the lowest fraction. The quartile-level patterns (Tables 1, 3) are stark enough to trust qualitatively; finer claims (e.g. exact convergence-rate percentages) carry real sampling uncertainty at this n .
- Quality scalars (Table 2, bottom two rows) are averaged only over *converged* units at each fraction – as convergence rate drops, this average is increasingly selected on a shrinking, potentially biased subset (as few as 3 units at fraction 0.25), not a random sample of “quality regardless of outcome.”
- This report covers FastMspec only, the technique the sweep was designed around; single-taper’s own Round 1 result (0/380 converged, a real mechanistic result) is separate context, not re-evaluated here.